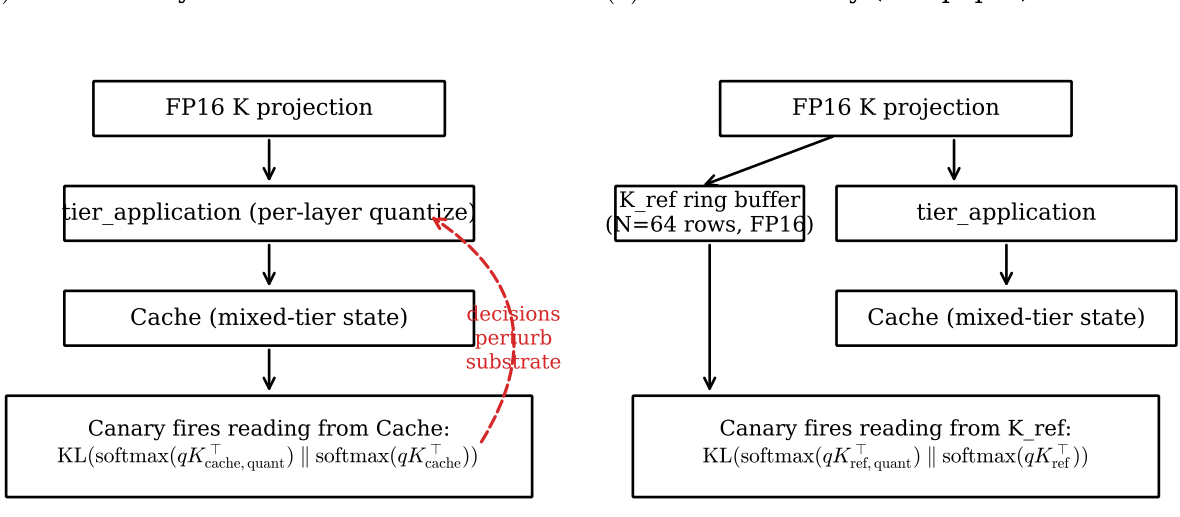
(a) naive canary
FP16 K projection
tier_application (per-layer quantize)
Cache (mixed-tier state)
decisions perturb substrate
Canary fires reading from Cache:
KL(softmax(qK⊤cache, quant) ‖ softmax(qK⊤cache))
(b) absolute canary (this paper)
FP16 K projection
K_ref ring buffer (N=64 rows, FP16)
tier_application
Cache (mixed-tier state)
Canary fires reading from K_ref:
KL(softmax(qK⊤ref, quant) ‖ softmax(qK⊤ref))
K_ref independent of cache state — no feedback path